

DIGITAL PRODUCT DESIGN

Lecture 4: Eliciting Consumer Preferences

Alex Moehring

Overview

- Model of consumer choice
- User studies to measure preferences
- Conjoint experiments to generate choice data
- Interpreting conjoint experiments

INTRODUCTION TO USER PREFERENCES

Introduction to Preferences

- Economics models individual choice as follows
 - Consumer must choose an alternative x from a set X
 - Consumer chooses alternative that maximizes utility $u(x)$ which maps alternatives to “utils” (real numbers)
- Preferences describe the parameters of the utility function
- If we know preferences, we can use this information to make better decisions

Example

- $X = \{apple, banana\}$
- $u(apple) = 7, u(banana) = 12$
- What fruit would you choose?

Product Features

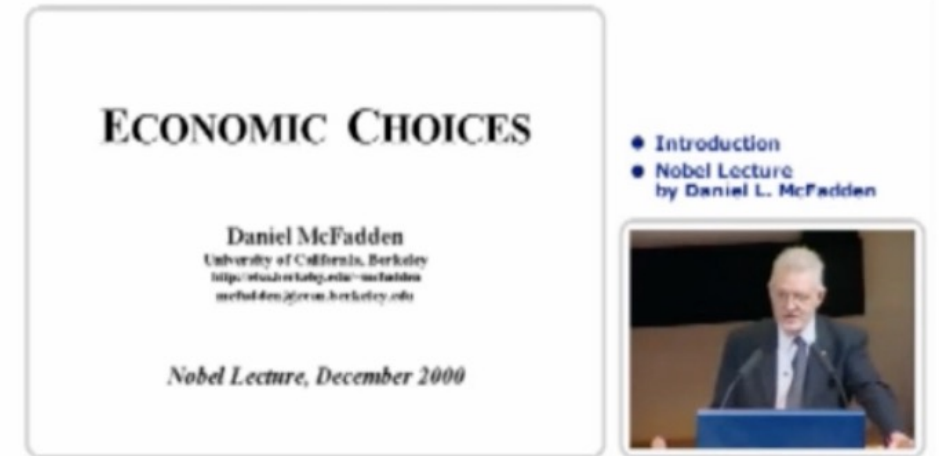
- Often useful to model utility as a function of product features
 - Each alternative x_i is a bundle of features $(x_{i1}, \dots, x_{iK})$
 - Example: Car can be described by size, engine, number of seats, number of cupholders, quality of infotainment system ...
- Model utility as a linear function of parameters
$$u(x_i) = \beta_1 x_{i1} + \dots + \beta_K x_{iK} + \epsilon_i$$
- Parameters $\beta_1, \dots, \beta_K$ are called preferences
 - Capture how much each feature impacts utility
- Question: Why can we not estimate this model using linear regression?

Product Features, Continued

- We do not observe $u(x)$!
- We often observe individual choices
- Can we estimate preferences using only data on choices?
 - Turns out the answer is yes!

Daniel L. McFadden Prize Lecture

Economic Choices



Multinomial Logit Model

- If ϵ_i has a particular distribution, then we can write choice probabilities in closed form
 - We then find the preference parameters that match predicted choice probabilities with actual choice probabilities
- Multinomial logit model allows us to take data on consumer choices and learn preferences!
 - If we have historical data on consumer choices, we can estimate preferences
 - Can also use survey methods to generate own data

USER STUDIES TO LEARN CUSTOMER PREFERENCES

Types of User Studies

- Interviews: One-on-one interview with potential customers
- Focus group: Moderator facilitates discussion between many customers
- Observe product in use

Benefits of User Studies

- Open ended and can be run anytime in the development process
- Can help you generate new features / solutions to explore
- Get to observe users using product and understand thinking

Downsides of User Studies

- Consumers have a preference for many features
 - Hard to know how strong that preference is from these studies
- Some risk of survey demand bias
 - People like to tell you things you want to hear

CONJOINT SURVEYS

Conjoint Overview

- We want to measure consumer preferences from choices
- Idea is to get survey responses to make choices in a simulated environment
- Pros:
 - Easy to set up and run without building prototype
 - Better than asking consumers “how much do you like X feature”
- Cons
 - Simulated choices are worse than actual choice data

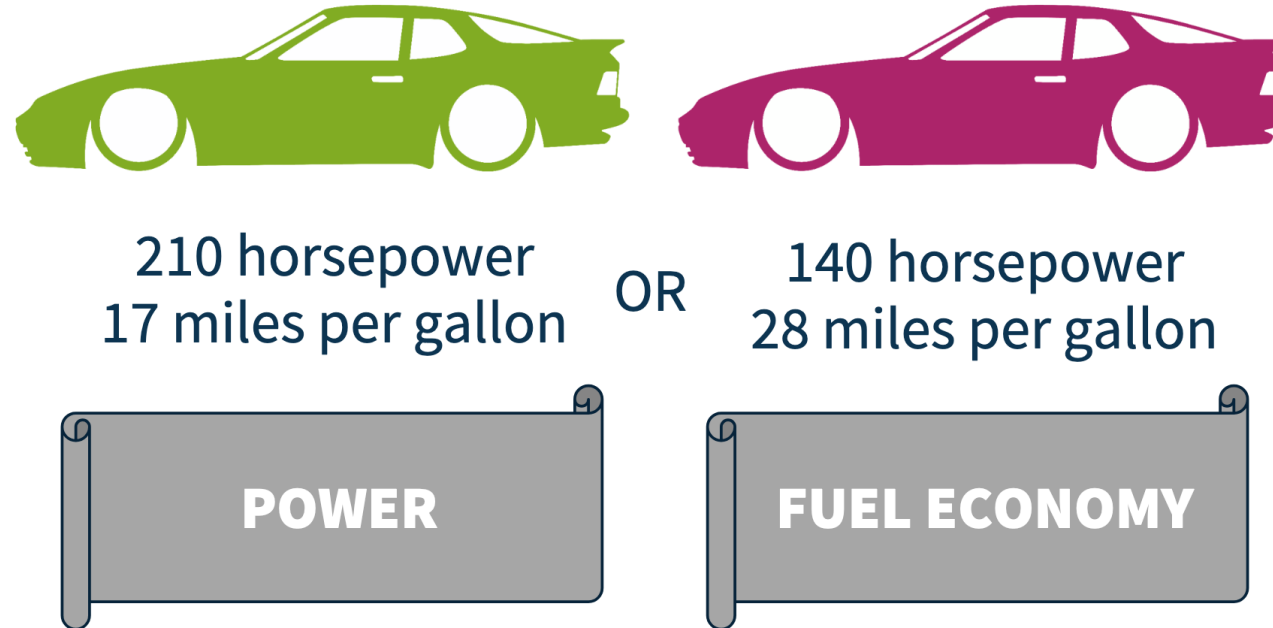
Conjoint Overview, Continued

- With data on choices, we let software to estimate preferences
- This can help us:
 - Understand how much consumers value different attributes
 - Prioritize features in the development process
 - Understand how to price our offering
 - Understand how to segment our customers

Mechanics of Running a Conjoint

- Use online software that offers “Choice Based Conjoint”
 - Sawtooth is a good option with a free tier
 - Conjointly also good but expensive
 - I have been told Qualtrics conjoint is not great
- Specify product features and levels (important to include price)
- Software automates randomization and data collection
- Recruit survey participants (or use LLMs)
- Software analyzes results

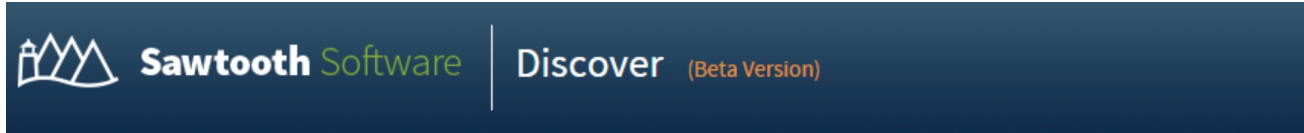
Participants Consider Several Bundles of Features



When respondents make difficult tradeoffs, we learn what they value!

Source: Sawtooth Software

Running a Conjoint using Sawtooth



Log into your Sawtooth Software Discover Account

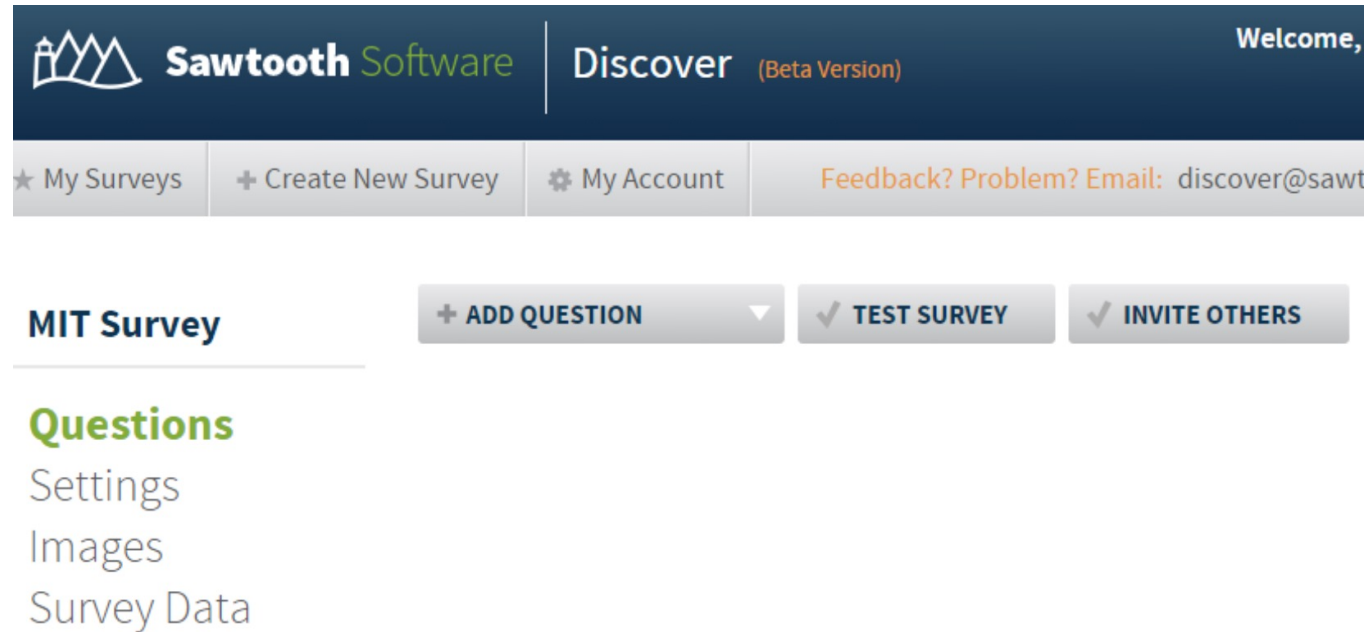
Email

Password

[Log In](#)

[Create New Account](#)

Add Choice Based Conjoint (CBC) Question



The screenshot shows the Sawtooth Software Discover interface. The top navigation bar includes the Sawtooth Software logo, the text "Discover (Beta Version)", and a "Welcome," message. Below this is a secondary navigation bar with links: "★ My Surveys", "+ Create New Survey", "⚙ My Account", and "Feedback? Problem? Email: discover@sawtooth.com". The main content area features a header for "MIT Survey" with three action buttons: "+ ADD QUESTION", "✓ TEST SURVEY", and "✓ INVITE OTHERS". A left sidebar contains a list of options: "Questions" (highlighted in green), "Settings", "Images", and "Survey Data".

Sample Survey

Imagine you were choosing a camera. You had done some research and are choosing between different models. A DSLR lens helps with the image quality. The number of megabytes affect the resolution of the photos you take. Which of these models would you choose?

1 / 6

Resolution	16 Mega Pixels	20 Mega Pixels	20 Mega Pixels	NONE: I wouldn't choose any of these.
DSLR Lens	Yes	Yes	No	
Price	\$200	\$400	\$200	
	☐	☒	☐	☐



0%  100%

Utility Analysis

Utilities Report					Show Chart
Total Respondents: 63					Download Individual Results
Resolution					
Levels	Utilities	Standard Deviations	Lower 95% CI	Upper 95% CI	
12 Mega Pixels	-37.09	25.30	-43.34	-30.84	
16 Mega Pixels	5.82	14.73	2.18	9.45	
20 Mega Pixels	31.28	24.17	25.31	37.25	
DSLR Lens					
Levels	Utilities	Standard Deviations	Lower 95% CI	Upper 95% CI	
Yes	63.55	26.34	57.05	70.06	
No	-63.55	26.34	-70.06	-57.05	
Price					
Levels	Utilities	Standard Deviations	Lower 95% CI	Upper 95% CI	
\$200	51.59	29.69	44.26	58.92	
\$400	1.35	20.42	-3.69	6.40	
\$600	-52.94	29.13	-60.14	-45.75	
None					
	Utilities	Standard Deviations	Lower 95% CI	Upper 95% CI	
None	-15.28	55.83	-29.07	-1.49	

Calculating Dollar Value of Features

- Must have price as an attribute in the conjoint
- Calculate the dollar value of a util
- Calculate the (relative) dollar value of features
 - If feature is color: green or red we can calculate dollar value of red relative to green, not the dollar value of a color
 - If feature is present / not present, then we can calculate dollar value of feature

Calculating the Dollar-Value of a “Util”

Price

Levels	Utilities	Standard Deviations	Lower 95% CI	Upper 95% CI
\$200	51.59	29.69	44.26	58.92
\$400	1.35	20.42	-3.69	6.40
\$600	-52.94	29.13	-60.14	-45.75

- First focus on the Price attribute
- Changing price from \$600 to \$200 increases utility by 104.5 utils
- \$400 off price increases utility by 104.5
 - \$1 = 0.26 utils
 - 1 util = \$3.85

Calculating the Dollar-Value of DSLR Feature

DSLR Lens

Levels	Utilities	Standard Deviations	Lower 95% CI	Upper 95% CI
Yes	63.55	26.34	57.05	70.06
No	-63.55	26.34	-70.06	-57.05

- We know 1util=\$3.85
- Adding SLR increases utility by 127 utils
- $127 \text{ utils} * 3.85 \text{ dollars per util} = \488 for SLR

Calculating the Dollar-Value of 1MP

Resolution

Levels	Utilities	Standard Deviations	Lower 95% CI	Upper 95% CI
12 Mega Pixels	-37.09	25.30	-43.34	-30.84
16 Mega Pixels	5.82	14.73	2.18	9.45
20 Mega Pixels	31.28	24.17	25.31	37.25

- We know 1util=\$3.85
- Adding 8MPs increases utility by 68 utils
- Adding 1 MP increases utility by $68/8 = 8.5$ utils
- $8.5 \text{ utils} * 3.85 \text{ dollars per util} = \38 per MP

Practice Interpreting Conjoint Results

- Company trying to sell jogging pants subscription service
 - Get new athletic wear monthly
- Calculate dollar value of a util
- Calculate dollar value of customized fit

Feature	Level	Utils
Monthly Price	\$100	200
Monthly Price	\$150	0
Monthly Price	\$200	-200
Customized Fit	Yes	50
Non-Customized Fit	No	-50
Number of Jogging Pants	1	-30
Number of Jogging Pants	2	10
Number of Jogging Pants	3	20

Practice Interpreting Conjoint Results, Continued

- Calculate dollar value of extra pairs of pants
 - Is this linear?
- What should we do given this information?

Feature	Level	Utils
Monthly Price	\$100	200
Monthly Price	\$150	0
Monthly Price	\$200	-200
Customized Fit	Yes	50
Non-Customized Fit	No	-50
Number of Jogging Pants	1	-30
Number of Jogging Pants	2	10
Number of Jogging Pants	3	20

Dos and Don'ts of Conjoint Analysis

- Do:
 - Test a broad range of prices
 - If we test \$19.99 and \$20, we will be extrapolating based off left digit biases
 - Better to choose \$10, \$20, \$30
 - Measure demographic information for customer segmentation
 - Do certain types of people value different features differently
 - Try to find respondents who are representative of your market

Dos and Don'ts of Conjoint Analysis, Continued

- Don't:
 - use conjoint if survey-takers are ignorant about the features
 - use conjoint if price is a signal of the quality
 - use conjoint with highly correlated features
 - E.g. if product is car, cannot have both features:
 - Type: coupe / mini-van
 - Number of seats: 2, 4, 8
 - calculate dollar value of utils if price relationship is highly non-linear

Final Points

- Don't include too many attributes or levels
 - Overloads users and requires a lot more responses
 - Features: 3-4 max
 - Levels: No more than 3
- Number of survey takers:
$$\frac{\#survey\ takers \times \#tasks \times \#concepts}{largest\ number\ of\ attributes} \geq 500$$
- Tasks: Number of choices a person makes (<10)
- Concepts: Number of options people choose between (<3)

FINDING PARTICIPANTS

Finding Participants for Surveys

- Actual customers
- Social media
- Paid panels
- Simulated AI respondents

Actual Customers

- Pro:
 - Representative of your customer segment
- Cons:
 - May get confused seeing features that don't currently exist
 - Cannot send person many surveys, risk of exhausting the pool

Social Media

- Two strategies:
 - Post your survey, try to get your community to participate and share
 - Called snow-ball sampling
 - Paid advertising
- Pro:
 - Often cheaper than paid panels (sometimes free!)
 - Gets feedback from most engaged customers
- Cons:
 - Often a non-representative sample

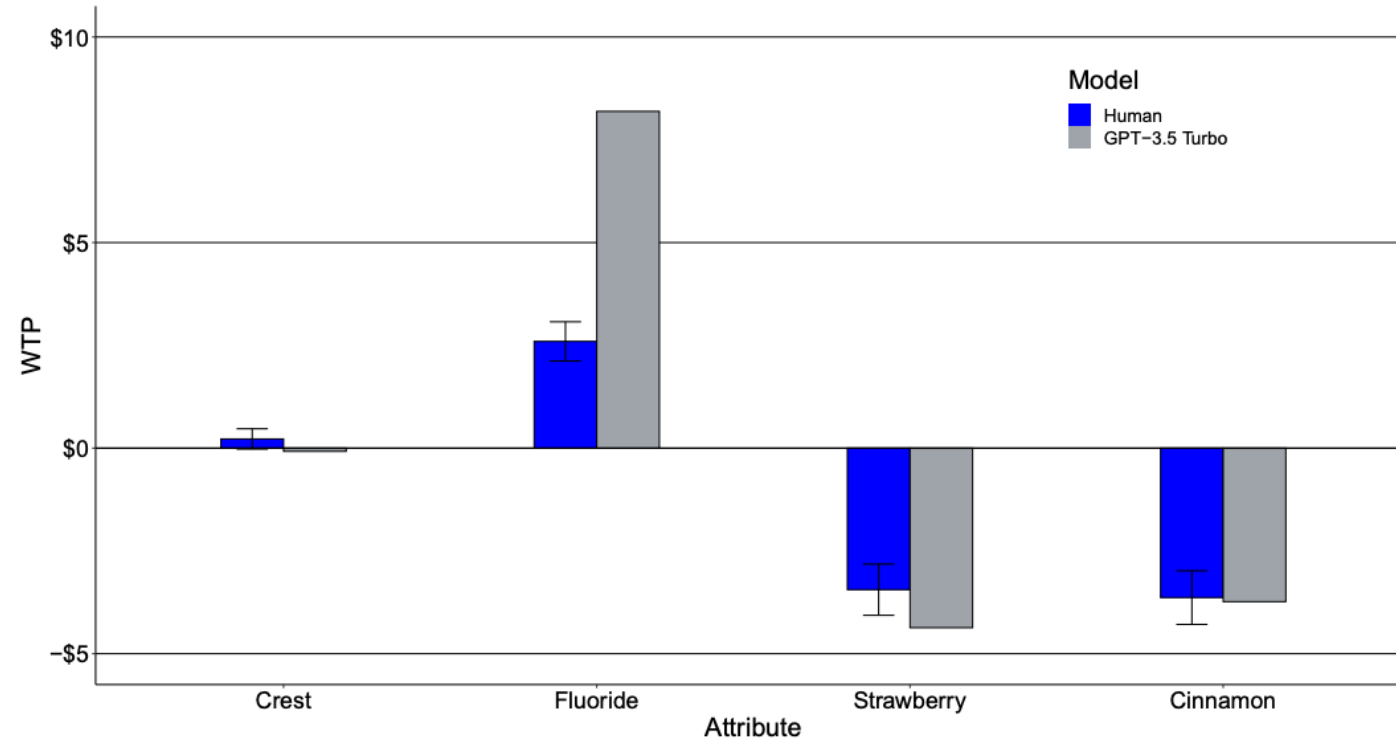
Paid Panels

- Companies pay individuals to take surveys / polls
 - Examples: Harris, Nielson, Yougov, ...
- Pros
 - Can get samples representative on many important dimensions
 - Some quality control exists
- Cons
 - Can be very expensive (think >\$10 per response)
 - Risk of using “professional” survey takers

- Ask ChatGPT or your LLM of choice to choose between alternatives
- Pros
 - Inexpensive to collect a lot of data
- Cons
 - Lack of understanding of when these mimic human responses
 - Evidence that responses are sensitive to prompt

LLMs vs Human Participants

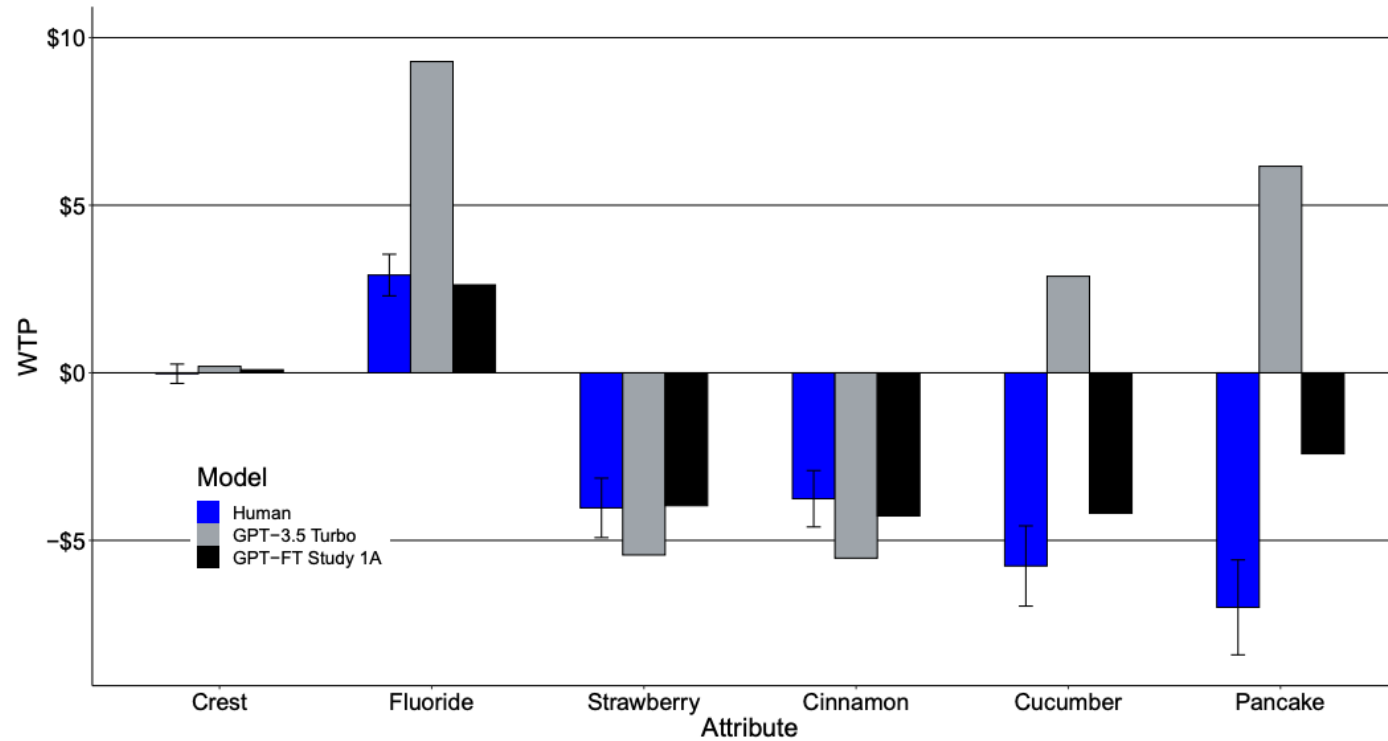
Figure 2: Results of Toothpaste Studies



(a) Study 1A: Toothpaste

Source: Brand et al (2024)

LLMs vs Human Participants, Continued



(b) Study 1B: Toothpaste with New Flavors

Source: Brand et al (2024)

Takeaway

- LLMs are a great way to do a rough cut of your survey
- There is still a lot to learn about when these results are reliable
 - Should we invest in pancake toothpaste?
 - Results are better with well-known products / features
- Some evidence that fine-tuning models can improve performance
- Risky to rely only on these models at this point

QUESTIONS

Purpose of Conjoint Surveys

- Conjoint surveys give us a method to measure the value of features
 - Can do this for different customer segments
- Helps us:
 - Prioritize development of most valuable features
 - Price our offering and segment the market

Next Time

- No Class Thursday
- Each group signs up for a project check-in meeting (~10 minutes)
- <https://calendar.app.google/er8YzgovSxjgj1JM8>